JAE YOUNG JANG

Ph.D. Student, Future Space Lab
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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) **Mar. 2025 – Present**
Ph.D., Department of Civil and Environmental Engineering Daejeon, Korea
Graduate School of Metaverse

Korea Advanced Institute of Science and Technology (KAIST) **Mar. 2023 – Feb. 2025**
M.S., Graduate School of Metaverse Daejeon, Korea
Thesis: *Complex Human Activity Recognition in Residential Spaces with Sector-Level Approach*
(Advisor: Seung Hyun Cha)

Seoul National University of Science and Technology **Mar. 2017 – Feb. 2023**
B.S., Department of Computer Science and Engineering Seoul, Korea
GPA: 4.18/4.5 (7th place out of 150), Major GPA: 4.39/4.5

RESEARCH PROFILE

My research focuses on **context-aware smart home automation**, integrating **contextual information** about users and their environments to enable both **reactive and proactive control**.

RESEARCH FUNDING

National Research Foundation of Korea (NRF) **2025 – 2027**
Ph.D. Student Research Encouragement Grant

- Role: Principal Investigator
- Title: Sector-level Spatial Segmentation and Multimodal Sensor Fusion for Complex Human Activity Recognition in Residential Environment

PUBLICATIONS

International Journals

- **Jang, J. Y.**, & Cha, S. H. (2026). Context-integrated complex activity recognition with sector-level location encoding in residential environments. *Advanced Engineering Informatics*, *76*, 105043. **[SCIE, Q1, 2.0%]**
- Seol, H., Yang, J., **Jang, J. Y.**, Choi, D., Han, G., & Cha, S. H. (2026). Development of a Physical Security Risk Matrix Integrating Heterogeneous Sensor Data and Spatiotemporal Context: Expert-Derived Parameters and Scenario-Based Validation. *Architectural Research*, *28*(1), 28.
- Song, Y., **Jang, J. Y.**, Han, G., & Cha, S. H., A Computational Framework for Integrating Activity-Based Space Utilization Optimization and Office Layout Generation. *Journal of Computational Design and Engineering* (Under Review).

International Conferences

- Song, Y., **Jang, J. Y.**, Eum, Y., & CHA, S. H. (2024). Space Utilization Based Framework For Designing Office Buildings For Efficient Space Use. In *International Conference of Smart and Sustainable Built Environment (SASBE 2024)*. Auckland University of Technology.
- Song, Y., **Jang, J. Y.**, Eum, Y., & Cha, S. H. (2024, November). A Conceptual Framework of Optimized Office Floor Plan Generation Based on Space Utilization Prediction. In *The International Conference of Sustainable Development and Smart Built Environments* (pp. 34–43). Singapore: Springer Nature Singapore.

International Posters

- Bae, S. A., Eum, Y., Kim, H. W., Kim, Y., Jeon, D., **Jang, J. Y.**, & Cha, S. H. (2025, October). Adaptive Virtual Reality Space Design Based on Real-Time User Emotion. In *2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 63–66). IEEE.

Domestic Journals

- Song, Y., **Jang, J. Y.**, & Cha, S. H. (2024). Development of Methodology for Automated Office Room Generation Based on Space Utilization. *Journal of KIBIM*, *14*(3), 1–12.

Domestic Conferences

- Yang, J., **Jang, J. Y.**, Seol, H., Kwon, D., Choi, D., Cheon, H.-C., Han, G., & Cha, S. H. (2026). An AI-City Integrated Physical Security Monitoring Framework: VLM Anomaly Detection, Heterogeneous Data Fusion-Based Risk Inference, and LLM-Based SOP Automation. *Proceedings of the 2026 Summer Conference of the Society for Computational Design and Engineering*.
- **Jang, J. Y.**, Kim, H., & Cha, S. H. (2024). Occupant Activity Recognition in Smart Homes - The Need for Expanded Research on Sensor-Based Human Activity Recognition. *Proceedings of the Spring Annual Conference of the Architectural Institute of Korea*, *44*(1), 482–485.
- **Jang, J. Y.**, Song, Y., & Cha, S. H. (2023). Automatic floorplan generation considering energy efficiency - Focused on the Energy Performance Prediction. *Proceedings of the Autumn Annual Conference of the Architectural Institute of Korea*, *43*(2), 501–504.

Domestic Posters

- Jung, K., Park, J., **Jang, J.**, Kim, H. I., & Woo, W. (2024). TeleAssiBot: Dual Secretary Assistive Robot. *Proceedings of the HCI Korea Conference*, 1004.

RESEARCH PROJECTS

- Sector-level Spatial Segmentation and Multimodal Sensor Fusion for Complex Human Activity Recognition in Residential Environment, *PI*, National Research Foundation **2025–2027**
- Development and Commercialization of AI-based Next-Generation Physical Security Integrated Monitoring System, *Researcher*, Korea Internet & Security Agency **2026**
- Development of Core Technologies for the Design, Construction, and Lifecycle Management of Robot-Adaptive Buildings, *Researcher*, Korea Agency for Infrastructure Technology Advancement **2025–2029**
- AI-Based Responsive-Transformative Spatial Technology for Future Space Innovation, *Researcher*, National Research Foundation **2025–2028**
- AR-AI-Cognition-Architectural Research for Transdimensional Interaction, *Researcher*, National Research Foundation **2024–2025**
- Real-time XR Interface Technology for Environmental Adaptation, *Researcher*, Institute for Information & Communications Technology Planning & Evaluation **2024–2025**
- Establishment of an International Network for Generative Agent Simulation-based Architecture Design Standard, *Researcher*, National Research Foundation **2024–2025**
- Development of Pre-Construction Spatial Evaluation System based on Multimodal Virtual Reality, *Researcher*, National Research Foundation **2023–2028**
- Space Design Framework for the Metaverse, *Researcher*, National Research Foundation **2022–2025**
- Development of IoT/AI based building envelope platform responsive to air-borne disease (COVID-19) and fire disasters, *Researcher*, National Research Foundation **2022–2027**
- Development of Artificial Intelligence Based Architectural Design Automation Technologies, *Researcher*, Korea Agency for Infrastructure Technology Advancement **2021–2025**

AWARDS

- **2024** HCI Korea Best Poster Award
- **2023** AIK (Architectural Institute of Korea) Best Paper Presentation Award
- **2021** Big Data Platform Utilization Idea Contest (Encouragement Prize)